



STAR PATTERN PROGRAMS

Pyramid Pattern

```
In [18]: rows=5
for i in range (1, rows + 1):
    print(' ' * (rows - i) +
          '*' * (2 * i - 1))
```

```
  *
 ***
*****
*****
*****
```

Downward Pyramid Pattern

```
In [21]: rows = 5
for i in range(rows, 0, -1):
    print(' ' * (rows-i) +
          '*' * (2 * i - 1))
```

```
*****
*****
*****
***
*
```

Double Pyramid

```
In [26]: rows = 5
# Upper half
for i in range (1, rows + 1):
    print(' ' * (rows - i) +
          '*' * (2 * i - 1))
# Lower half
for i in range(rows - 1, 0, -1) :
    print(' ' * (rows - i) +
          '*' * (2 * i - 1))
```

```
  *
 ***
*****
*****
*****
*****
*****
***
*
```

Butterfly Pattern

```
In [29]: rows = 5
# Upper part
for i in range(1, rows + 1):
    print('*' * i + ' ' * (2 * (rows - i)) + '*' * i)
# Lower part
for i in range (rows,0,-1):
    print ('*' * i + ' ' * (2 * (rows-i)) + '*' * i)
```

```
*      *
**     **
***    ***
****   ****
*****
*****
****   ****
***    ***
**     **
*      *
```

Rightward Triangle

```
In [1]: rows = 5
for i in range(1, rows + 1):
    print('*' * i)
```

```
*
**
***
****
*****
```

Leftward Triangle

```
In [6]: rows = 5
for i in range(1, rows + 1):
    print(' ' * (rows - i) + '*' * i)
```

```
    *
   **
  ***
 ****
*****
```

Downward Triangle

```
In [14]: rows = 5
for i in range(rows , 0, -1):
```

```
print('*' * i)
```

```
*****
****
***
**
*
```

Triangle

```
In [17]: rows = 5
for i in range (1, rows +1):
    if i == rows:
        print('*' * (2 * i - 1))
    else:
        print(' ' * (rows - i) +
              '*' + ' ' * (2 * i - 3) +
              ('*' if i > 1 else ''))
```

```
      *
     * *
    *  *
   *   *
  *    *
*****
```

Zero Pattern

```
In [20]: rows = 5
for i in range (rows):
    if i == 0 or i == rows - 1:
        print('*' * rows)
    else:
        print('*' + ' ' * (rows - 2) + '*')
```

```
*****
*   *
*   *
*   *
*****
```

Diamond Pattern

```
In [26]: rows = 5
# Upper part
for i in range (1, rows + 1):
    print(' ' * (rows - i) +
          '*' + ' ' * (2 * i - 3) +
          ('*' if i > 1 else ''))
# Lower part
for i in range (rows - 1, 0, -1):
```

```
print(' ' * (rows - i) + '*'
+ ' ' * (2 * i - 3) + ('*'
if i > 1 else ''))
```

```
*
* *
*  *
*    *
*      *
*        *
*          *
*            *
*              *
```

X Pattern

```
In [29]: rows= 5
for i in range(1, rows + 1):
    for j in range(1, rows + 1):
        if i == j or i + j == rows + 1:
            print('*', end='')
        else:
            print(' ', end='')
    print ()
```

```
*  *
* *
*
* *
*  *
```

Star Pattern

```
In [40]: import math

def main():
    q = math.tan(math.pi * 0.4)
    w = math.tan(math.pi * 0.2)

    n = float(input("Enter the size:\n"))

    for j in range(math.ceil(n * q), -1, -1):
        for i in range(-math.ceil(0.55 * n * q / w - n), math.ceil(0.55 * n * q / w - n) + 1):
            if (
                ((j <= 0.55 * n * q and j >= (i + n) * w and j >= (n - i) * w)
                or (j >= (i + n) * w and j <= (i + n) * q and j <= (n - i) * q)
                or (j <= (n - i) * q and j >= (n - i) * w and j <= (i + n) * w)
            ):
                print("*", end=" ")
            else:
                print(" ", end=" ")
        print()
```

```
print()

if __name__ == "__main__":
    main()
```

[illegible]